Problem

Problem · The affine cipher. The letters A to Z are numbered from 0 to 25. To encrypt a letter with number x, compute the remainder y of $7x + 3$ in the division by 26, and replace the letter by the letter with number y.

1. Encrypt the word EXPERT.
2. Prove that $y \equiv 7x + 3 \pmod{26}$ is equivalent to $x \equiv 15y + 7 \pmod{26}$.
3. Decrypt the message TDNZZ.
4. Replace 7 by a in the key. Prove that if $\gcd(a, 26) \neq 1$, two different letters may be encrypted as the same letter. How many values of $a \in \{1, \ldots, 25\}$ give a cipher that can be decrypted?